Panosec Tablet 20mg

Name of Product
Pantoprazole Gastro-Resistance Tablet 20mg

Name and Strength of Active Ingredient
Name : Pantoprazole Sodium Sesquihydrate
Strength : 20mg

Route of Administration: Oral

Product Description
Yellow to pale yellow, oval, biconvex gastro resistant tablets imprinted'H125' on one side with black ink and plain on the other side.

Pharmacodynamics
Mechanism of action
Pantoprazole is a substituted benzimidazole which inhibits the secretion of hydrochloric acid in the stomach by specific blockade of the proton pumps of the parietal cells. Pantoprazole is converted to its active form in the acidic environment in the parietal cells where it inhibits the H+, K+-ATPase enzyme, i.e. the final stage in the production of hydrochloric acid in the stomach. The inhibition is dose- dependent and affects both basal and stimulated acid secretion. In most patients, freedom from symptoms is achieved within 2 weeks. As with other proton pump inhibitors and H2 receptor inhibitors, treatment with pantoprazole reduces acidity in the stomach and thereby increases gastrin in proportion to the reduction in acidity. The increase in gastrin is reversible. Since pantoprazole binds to the enzyme distal to the cell receptor level, it can inhibit hydrochloric acid secretion independently of stimulation by other substances (acetylcholine, histamine, gastrin). The effect is the same whether the product is given orally or intravenously. The fasting gastrin values increase under pantoprazole. On short-term use, in most cases they do not exceed the upper limit of normal. During long-term treatment, gastrin levels double in most cases. An excessive increase, however, occurs only in isolated cases. As a result, a mild to moderate increase in the number of specific endocrine (ECL) cells in the stomach is observed in a minority of cases during long term treatment (simple to adenomatoid hyperplasia).
During treatment with antisecretory medicinal products, serum gastrin increases in response to the decreased gastric acidity. The increased CgA level may interfere with investigations for neuroendocrine tumours.
Available published evidence suggests that proton pump inhibitors should be discontinued between 5 days and 2 weeks prior to CgA measurements. This is to allow CgA levels that might be spuriously elevated following PPI treatment to return to reference range.

Pharmacokinetics
Absorption
Pantoprazole is rapidly absorbed and the maximal plasma concentration is achieved even after one single 20 mg oral dose. On average at about 2.0 h - 2.5 h p.a. the maximum serum concentrations of about 1-1.5 µg/ml are achieved, and these values remain constant after multiple administration. Pharmacokinetics do not vary after single or repeated administration. In the dose range of 10 to 80 mg, the plasma kinetics of pantoprazole are linear after both oral and intravenous administration.
The absolute bioavailability from the tablet was found to be about 77 %. Concomitant intake of food had no influence on AUC, maximum serum concentration and thus bioavailability. Only the variability of the lag- time will be increased by concomitant food intake.
Distribution
Pantoprazole's serum protein binding is about 98 %. Volume of distribution is about 0.15 l/kg.
Biotransformation
The substance is almost exclusively metabolized in the liver. The main metabolic pathway is demethylation by CYP2C19 with subsequent sulphate conjugation, other metabolic pathway include oxidation by CYP3A4.
Elimination
Terminal half-life is about 1 hour and clearance is about 0.1 l/h/kg. There were a few cases of subjects with delayed elimination. Because of the specific binding of pantoprazole to the proton pumps of the parietal cell the elimination half-life does not correlate with the much longer duration of action (inhibition of acid secretion).
Renal elimination represents the major route of excretion (about 80 %) for the metabolites of pantoprazole, the rest is excreted with the faeces. The main metabolite in both the serum and urine is desmethylpantoprazole which is conjugated with sulphate. The half-life of the main metabolite (about 1.5 hours) is not much longer than that of pantoprazole.
Children
Following administration of single oral doses of 20 or 40 mg pantoprazole to children aged 5 - 16 years. AUC and Cmax were in the range of corresponding values in adults.
Following administration of single i.v. doses of 0.8 or 1.6 mg/kg pantoprazole to children aged 2 – 16 years, there was no significant association between pantoprazole clearance and age or weight. AUC and volume of distribution were in accordance with data from adults.

Indications
For the treatment of mild gastroesophageal reflux disease and associated symptoms (e.g. heartburn, acid regurgitation, pain on swallowing). For long term management and prevention of relapse in reflux oesophagitis. Prevention of gastrointestinal ulcers induced by non-selective non-steroidal anti- inflammatory drugs (NSAID) in patients at risk with a need for continuous NSAID treatment.

Recommended dose
Dosage and method of administration of Panosec Tablet 20mg unless otherwise prescribed, the following oral doses apply. Please follow these instructions, as otherwise there is the risk that Panosec Tablet 20mg may not act properly.

Adults and adolescents 12 years of age and above:
Mild reflux disease and associated symptoms (e.g. heartburn, acid regurgitation, pain on swallowing): the recommended oral dosage is one Panosec Tablet 20mg per day.
Long-term management and prevention of relapse in reflux oesophagitis: Maintenance dose of one Panosec Tablet 20mg per day is recommended, increasing to 2 tablets of Panosec Tablet 20mg per day if a relapse occur. Panosec Tablet 40mg is available for this case. After healing of the relapse, the dosage can be reduced again to one Panosec Tablet 20mg per day.

Adults:
Prevention of gastroduodenal ulcers induced by non-selective non-steroidal anti-inflammatory drugs (NSAIDs) in patients at risk with a need for continuous NSAID treatment: The recommended dose is one Panosec Tablet 20mg perday.

Children below 12 years of age:
Panosec Tablet 20mg is not recommended for use in children below 12 years of age due to limited data in this age group.
Note: A daily dose of 20mg pantoprazole should not be exceeded in patients with severe liver impairment. No dose adjustment is necessary in elderly patients of in those with impaired renal function.

The use of pantoprazole 20mg as a preventive of gastroduodenal ulcers induced by non-selective non-steroidal anti-inflammatory drugs (NSAIDs) should be restricted to patients who require continued NSAID treatment and have an increased risk to develop gastrointestinal complications. The increased risk should be assessed according to individual risk factors, e.g. high age (> 65 years), history of gastric or duodenal ulcer or upper gastrointestinal bleeding.

Type and duration of treatment:
Mild reflux disease and associated symptoms (e.g. heartburn, acid regurgitation, pain on swallowing): the symptom relief is generally accomplished within 2 - 4 weeks, and a 4 weeks treatment period is usually required for healing of associate oesophagitis. If this is not sufficient, healing will usually be achieved within a further 4 weeks. When symptoms relief has been achieved, reoccurring symptoms can be controlled using an on-demand regimen of Panosec Tablet 20mg once daily, when required. A switch to continuous therapy may be considered in case satisfactory symptom control cannot be maintained with on-demand treatment.
Long term management and prevention of relapse in reflux oesophagitis: a treatment period of 1 year should be exceeded only after careful consideration of the benefit/risk ratio, as drug safety over several years is not sufficiently established.

Instructions for use/ handling: Panosec Tablet 20 mg must not be chewed or crushed and must be swallowed whole with water.

Contraindications
Panosec Tablet 20mg must not be used in cases of known hypersensitivity to the active ingredient or/and any of the other constituents of Panosec Tablet 20mg. Pantoprazole, like other PPIs, should not be co-administered with atazanavir (see section Interactions).

Warnings and precautions
Hepatic Impairment
In patients with severe liver impairment the liver enzymes should be monitored regularly during treatment with pantoprazole, particularly on long-term use. In the case of a rise of the liver enzymes the treatment should be discontinued.

Co-administration with NSAIDs
The use of Pantoprazole 20 mg as a preventive of gastroduodenal ulcers induced by non-selective non- steroidal anti-inflammatory drugs (NSAIDs) should be restricted to patients who require continued NSAID treatment and have an increased risk to develop gastrointestinal complications. The increased risk should be assessed according to individual risk factors, e.g. high age (>65 years), history of gastric or duodenal ulcer or upper gastrointestinal bleeding.

Gastric Malignancy
Symptomatic response to pantoprazole may mask the symptoms of gastric malignancy and may delay diagnosis. In the presence of any alarm symptom (e. g. significant unintentional weight loss, recurrent vomiting, dysphagia, haematemesis, anaemia or melaena) and when gastric ulcer is suspected or present, malignancy should be excluded.
Further investigation is to be considered if symptoms persist despite adequate treatment.

Co-administration with HIV protease inhibitors
Co-administration of pantoprazole is not recommended with HIV protease inhibitors for which absorption is dependent on acidic intragastric pH such as atazanavir, due to significant reduction in their bioavailability.

Vitamin B12 Deficiency
Daily treatment with any acid-suppressing medications over a long period of time (e.g. longer than 3 years) may lead to malabsorption of cyanocobalamin (vitamin B12) caused by hypo- or achlorhydria. Rare reports of cyanocobalamin deficiency occurring with acid-suppressing therapy have been reported in the literature. The diagnosis should be considered if clinical symptoms consistent with cyanocobalamin deficiency are observed.

Regular Surveillance
Patients on proton pump inhibitor treatment (particularly those treated for long term) should be kept under regular surveillance.

Gastrointestinal infections caused by bacteria
Treatment with Pantoprazole may lead to a slightly increased risk of gastrointestinal infections caused by bacteria such as Salmonella, Campylobacter or C. difficile.

Clostridium Difficile Diarrhea
Published observational studies suggest that PPI therapy may be associated with an increased risk of Clostridium difficile associated diarrhea, especially in hospitalized patients. This diagnosis should be considered for diarrhea that does not improve. Patients should use the lowest dose and shortest duration of PPI therapy appropriate to the condition being treated.

Hypomagnesaemia
Severe hypomagnesaemia has been reported in patients treated with PPIs like pantoprazole for at least three months, and in most cases for a year. Serious manifestations of hypomagnesaemia such as fatigue, tetany, delirium, convulsions, dizziness and ventricular arrhythmia can occur but they may begin insidiously and be overlooked. In most affected patients, hypomagnesaemia improved after magnesium replacement and discontinuation of the PPI.
For patients expected to be on prolonged treatment or who take PPIs with digoxin or drugs that may cause hypomagnesaemia (e.g., diuretics), health care professionals should consider measuring magnesium levels before starting PPI treatment and periodically during treatment.

Fractures
Proton pump inhibitors, especially if used in high doses and over long durations (>1 year), may modestly increase the risk of hip, wrist and spine fracture, predominantly in the elderly or in presence of other recognised risk factors. Observational studies suggest that proton pump inhibitors may increase the overall risk of fracture by 10-40%. Some of this increase may be due to other risk factors. Patients at risk of osteoporosis should receive care according to current clinical guidelines and they should have an adequate intake of vitamin D and calcium.

Subacute cutaneous lupus erythematosus (SCLE)
Proton pump inhibitors are associated with very infrequent cases of subacute cutaneous lupus erythematosus (SCLE). If lesions occur, especially in sun-exposed areas of the skin, and if accompanied by arthralgia, the patient should seek medical help promptly and the health care professional should consider stopping Pantoprazole. SCLE after previous treatment with a proton pump inhibitor may increase the risk of SCLE with other proton pump inhibitors.

Interference with Laboratory Tests
Increased Chromogranin A (CgA) level may interfere with investigations for neuroendocrine tumours. If the patient(s) are due to have a test on Chromogranin A level, Pantoprazole treatment should be stopped for at least 5 days before CgA measurements to avoid this interference (see section Pharmacodynamic). If CgA and gastrin levels have not returned to reference range after initial measurement, measurements should be repeated 14 days after cessation of proton pump inhibitor treatment.

Interaction with other medicaments

Medicinal products with pH-Dependent Absorption Pharmacokinetics

Because of profound and long-lasting inhibition of gastric acid secretion, pantoprazole may interfere with the absorption of other medicinal products where gastric pH is an important determinant of oral availability, e.g. some azole antifungals as ketoconazole, itraconazole, posaconazole and other medicine as erlotinib.

HIV protease inhibitors

Co-administration of pantoprazole is not recommended with HIV protease inhibitors for which absorption is dependent on acidic intragastric pH such as atazanavir due to significant reduction in their bioavailability.

If the combination of HIV protease inhibitors with a proton pump inhibitor is judged unavoidable, close clinical monitoring (e.g. virus load) is recommended. A pantoprazole dose of 20mg per day should not be exceeded. Dosage of the HIV protease inhibitor may need to be adjusted.

Coumarin anticoagulants (phenprocoumon or warfarin)

Co-administration of pantoprazole with warfarin or phenprocoumon did not affect the pharmacokinetics of warfarin, phenprocoumon or INR. However, there have been reports of increased INR and prothrombin time in patients receiving PPI's and warfarin or phenprocoumon concomitantly. Increases in INR and prothrombin time may lead to abnormal bleeding and even death. Patients treated with pantoprazole and warfarin or phenprocoumon may need to be monitored for increase in INR and prothrombin time.

Methotrexate

Concomitant use of high dose methotrexate (e.g. 300 mg) and proton-pump inhibitors has been reported to increase methotrexate levels in some patients. Therefore in settings where high-dose methotrexate is used, for example cancer and psoriasis, a temporary withdrawal of pantoprazole may need to be considered.

Medicinal products that inhibit or induce CYP2C19

Inhibitors of CYP2C19 such as flucoxamine could increase the systemic exposure of pantoprazole. A dose reduction may be considered for patients treated long-term with high doses of pantoprazole, or those with hepatic impairment.

Enzyme inducers affecting CYP2C19 and CYP3A4 such as rifampicin and St John's wort (Hypericum perforatum) may reduce the plasma concentrations of PPIs that are metabolised through these enzyme systems.

Pregnancy and Lactation

Pregnancy

As a precautionary measure, it is preferable to avoid the use of Pantoprazole during pregnancy.

Breast-feeding

Animal studies have shown excretion of pantoprazole in breast milk. Excretion into human milk has been reported. During breastfeeding, pantoprazole should only be used if the benefit to the woman outweighs the potential risk to the foetus or child.

Side Effects

a) Blood and lymphatic system disorders
Rare : Agranulocytosis
Very Rare : Thrombocytopenia; Leukopenia, Pancytopenia

b) Immune system disorders
Rare : Hypersensitivity (including anaphylactic reactions and anaphylactic shock)

c) Metabolism and nutrition disorders
Vitamin B12 deficiency
Rare : Hyperlipidaemias and lipid increases (triglycerides, cholesterol); Weight changes
Not known : Hyponatraemia; Hypomagnesaemia (see warnings and precautions). Hypocalcaemia in association with hypomagnesemia; Hypokalaemia

d) Psychiatric disorders
Uncommon : Sleep disorders
Rare : Depression (and all aggravations)
Very rare : Disorientation (and all aggravations)
Not known : Hallucination; Confusion (especially in pre-disposed patients, as well as the aggravation of these symptoms in case of pre-existence)

e) **Nervous system disorders**
Uncommon : Headache; Dizziness
Rare : Taste disorders
Not known : Paraesthesia

f) **Eye disorders**
Rare : Disturbances in vision / blurred vision

g) Gastrointestinal disorders
Common : Fundic gland polyps (benign)
Uncommon : Diarrhoea; Nausea /vomiting; Abdominal distension and bloating; Constipation; Dry mouth; Abdominal pain and discomfort
Not known: Microscopic colitis

h) Hepatobiliary disorders
Uncommon : Liver enzymes increased (transaminases, γ-GT)
Rare : Bilirubin increased
Not known : Hepatocellular injury; Jaundice; Hepatocellular failure

i) **Skin and sub-cutaneous tissue disorders**
Uncommon : Rash/exanthema/eruption; Pruritus
Rare : Urticaria; Angioedema
Not known : Stevens-Johnson syndrome; Lyell syndrome; Erythema multiforme; Photosensitivity; Subacute cutaneous lupus erythematosus (SCLE) (see warnings and precautions)

j) Musculoskeletal and connective tissue disorders
Uncommon : Fracture of the hip, wrist or spine (see warnings and precautions)
Rare : Arthralgia; Myalgia
Not known : Muscle spasm as a consequence of electrolyte disturbances

k) Renal and urinary disorders
Not known : Interstitial nephritis (with possible progression to renal failure)

l) Reproductive system and breast disorders
Rare : Gynaecomastia

m) General disorders and administration site conditions
Uncommon : Asthenia, fatigue and malaise
Rare : Body temperature increased; Oedema peripheral

n) Infections and Infestations
Clostridium diffi cile associated diarrhea

Symptoms and Treatment of Overdose

There are no known symptoms of over dosage in man. Systemic exposure with up to 240 mg administered intravenously over 2 minutes were well tolerated. As pantoprazole is extensively protein bound, it is not readily dialysable. In the case of overdose with clinical signs of intoxication, apart from symptomatic and supportive treatment, no specific therapeutic recommendations can be made.

Storage Conditions

Store below 30°C and protect from moisture.

Shelf Life

24 months

Pack Size

1 x 10’s, 3 x 10’s, 6 x 10’s, 9 x 10’s, 10 x 10’s and 50 x 10’s blister pack

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